

A
MINI PROJECT REPORT
ON
“FACE MASK DETECTOR”

Submitted in partial fulfillment of the requirement for the award of

THIRD YEAR

IN

ELECTRONICS AND TELECOMMUNICATION ENGINEERING

Submitted By

BAGUL ASHISH SURESH

GAIKWAD ROHIT DADAJI

MAHARSHI YADNYESH PRAMOD

Under the guidance of

PROF. P. D. LOKHANDE



Gokhale Education Society's

**R. H. Sapat College of Engineering,
Management Studies and Research, Nashik-5**
DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION
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ACADEMIC YEAR: 2021-2022



Gokhale Education Society's
**R. H. Sapat College of Engineering,
Management Studies and Research, Nashik-5**
DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION ENGINEERING

Certificate

This is to certify that, this mini project report entitled “Face Mask Detector” Submitted by Mr. Bagul Ashish Suresh Exam Seat No. T190693005, Mr. Gaikwad Rohit Dadaji Exam Seat No. T190693016, Mr. Maharshi Yadnyesh Pramod Exam Seat No. T190693030 for partial fulfillment of the requirement for the award of the degree THIRD YEAR in ELECTRONICS AND TELECOMMUNICATION ENGINEERING as laid down by SAVITRIBAI PHULE PUNE UNIVERSITY, Pune, is a record of their own work carried out by them under my supervision and guidance during year 2021 – 2022.

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PROF. P. D. LOKHANDE

PROJECT GUIDE

PROF.DR. S. P. AGNIHOTRI

HEAD OF DEPARTMENT

PROF.

SIGNATURE OF EXAMINER

DR. P. C. KULKARNI

PRINCIPAL

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**BAGUL ASHISH SURSH
GAIKWAD ROHIT DADAJI
MAHARSHI YADNYESH PRAMOD**

ABSTRACT

From more than two years, Coronavirus disease (COVID-19) is spreading all over the world. It is an infectious disease caused by SARS-COV-2 virus. It can affect any person who comes in contact of the virus, so it is very necessary to take all the precautions. It is very important to take care at public places where large number of people gather. Detecting whether person has worn a face mask or not in a crowd is a very difficult task. To make this easy, we need to take help of new technology using embedded systems and deep learning. Embedded systems are computer-based systems that have to perform a particular function. During this COVID time, we have to follow all the guidelines strictly. Using face recognition system in identifying face mask will be helpful for all of us will prevent us from the contact of virus. The project mainly focuses on reducing the manual presence in public places to check face masks. It ensures that the people and front-end workers can stay safe in these unprecedented times and ensures that everyone follows the guidelines which are provided by the government.

Keywords: *COVID-19, Embedded Systems, Deep Learning, Face Recognition System*

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LIST OF ABBREVIATIONS

1. COVID-19: Corona Virus Disease- 2019
2. WHO: World Health Organization
3. ESP: Espressif
4. CNN: Convolution Neural Network
5. YOLO: You Only Look Once
6. LED: Light-Emitting Diode
7. I/O: Input/Output
8. Tx: Transmitter
9. Rx: Receiver
10. TF: Trans-Flash
11. SD: Secure Digital
12. MP: Mega Pixel
13. QR: Quick Response
14. IDE: Integrated Development Environment
15. USB: Universal Serial Bus
16. IEEE: Institute of Electrical and Electronics Engineers

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1. INTRODUCTION

From March 2020, we are living in the pandemic situation due to corona virus. COVID-19 outbreak has created significant impact on everyone all of the world. According to WHO, coronavirus spreads mainly through the drops of saliva or liquid particles from the nose when infected person coughs or sneezes. So, it is important to wear a face mask. Using mask in public places help in preventing spread of COVID-19 disease.

It is tough to check each and every person in the crowd has worn a face mask or not. Also, it is very time consuming with requirement of human efforts. To overcome such problem, we have designed a system 'Face Mask Detector' which is used to monitor without human intervention. Hence it is used to detect whether the person is wearing a face mask or not.

The purpose of this project is face mask detection which results in reducing human efforts. With the use of deep learning, higher accuracies in face detection and face recognition can be acquired. This operation requires a special type of camera known as ESP32 CAM module which comes with ESP32 board.

This process is fast and there is no need of any physical interaction or any operator's supervision to start, operate or stop the system. It is completely autonomous and a type of smart work which can be done in pandemic situation. This can be used in many public places like malls, theatres, hospitals, schools, etc.

2. LITERATURE SURVEY

Various types of work are done for face mask detection in the pandemic situation to help the world to reduce efforts and stress. Xiaopeng Liu and Sisen Zhang [1] described the importance of using facemasks to avoid spreading coronavirus and they also discussed the origin of the coronavirus. Yukai Shi, Guanbin Li, Qingxing Cao, Keze Wang and Liang Lin [2] proposed a solution to produce a high-resolution image from a low-resolution image. Tanvir Ahmed, Prangon Das, Md. Firoj Ali, Md- Firoz Mahmud [3] developed a convolutional neural network design for face detection. CNN is a machine learning algorithm for machines to understand the features of the image.

Truong Quang Vinh, Nguyen Tran Ngoc Anh [4] discussed about the YOLOv3 algorithm and Haar Cascade classifier in their system for face mask detection. This algorithm is very complex and difficult to implement. Oladapo Ibitoye [5] reviewed convolutional neural network technique for masked face recognition. M Rohini, K.R Naveena, G Jothipriya, S Kameshwaran, M Jagadeeswari [6] discussed about the spread of corona virus using machine learning.

Wuttichai Vijitkunsawat, Peerasak Chantngarm [7] showed a study of performance of algorithms for face mask detection. A. A. Sawchuk [8] presented research on optical signal and image processing with analog and digital techniques. Lyla B. Das [9] showed integrated approach on embedded systems.

3. PROBLEM STATEMENT

COVID-19 is the greatest challenge in front of mankind. According to the WHO, COVID-19 disease spreads through corona virus from infected person to anyone by the droplets of saliva or liquid particles. So, at places where major number of people are gathering, it is mandatory to check everyone has wear a mask or not.

It is very challenging to do this process by any person where large number of people enter and pass. We can create solution of this problem using electronics system by detecting the mask availability of all the people individually.

4. OBJECTIVES OF PROJECT

To develop an Arduino based Face Mask Detector system which can check the person has worn a face mask or not.

To develop a system which shows a message turning on LED that a person has worn a face mask.

5. SPECIFICATIONS

5.1 Hardware Requirements

The 'Face Mask Detector' system is completely based on the function and features of ESP32 camera module. This camera module is required to scan the face of the person who is standing in front of the system. This module requires a source code to run and show output. We can provide a source code or a program to this module using Arduino Uno. To program ESP32 CAM module, a special type of circuit is required which is discussed further.

Arduino Uno is also required in this project. It is required to program ESP32 CAM module. It is also needed to give 3.3V supply to ESP32 CAM module using supply pins. Hence, Arduino Uno should be connected to power supply.

A LED lights are required i. e. green LED. This LED are connected to the ESP32 CAM module using I/O pins of it. First, the module will scan the face of person. If mask is present, a output signal will be given to him/her using green LED.

5.2 Software Requirements

To draw and design the circuits of the system and ESP32 CAM programming circuit, we used Fritzing software. It is an open-source hardware initiative that makes electronics accessible as a creative material for users.

The schematic circuit diagram and the simulation circuit of this project is drawn using Proteus software. The Proteus Design Suite is a proprietary software tool suite used primarily for electronic circuit automation.

We used Arduino IDE for the programming of ESP32 CAM module. It requires to add ESP32 library in the software. It is a software which is used to write and upload code to the Arduino boards.

6. BLOCK DIAGRAM AND ITS DESCRIPTION

6.1 Block Diagram

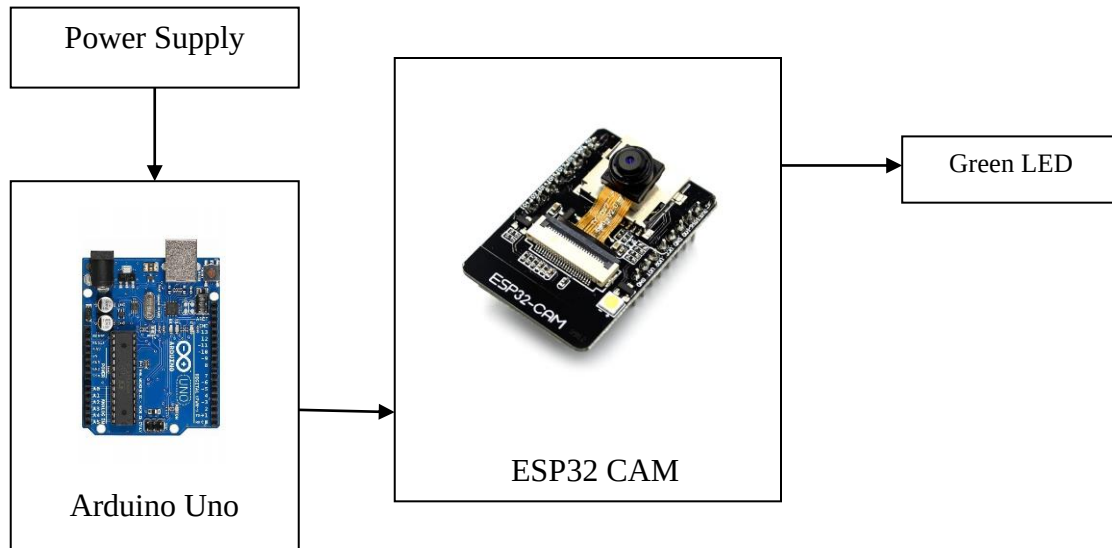


Fig. 6.1.1: Block Diagram

This block diagram consists of power supply, Arduino Uno, ESP32 CAM module, green LED and red LED. First, a regulated power supply should be given to the Arduino Uno using USB cable or adapter. Then 3.3V power supply should be given to ESP32 using Arduino Uno pins. Red and Green LED lights should be connected to the ESP32 CAM module to show the output by glowing.

6.2 Flowchart

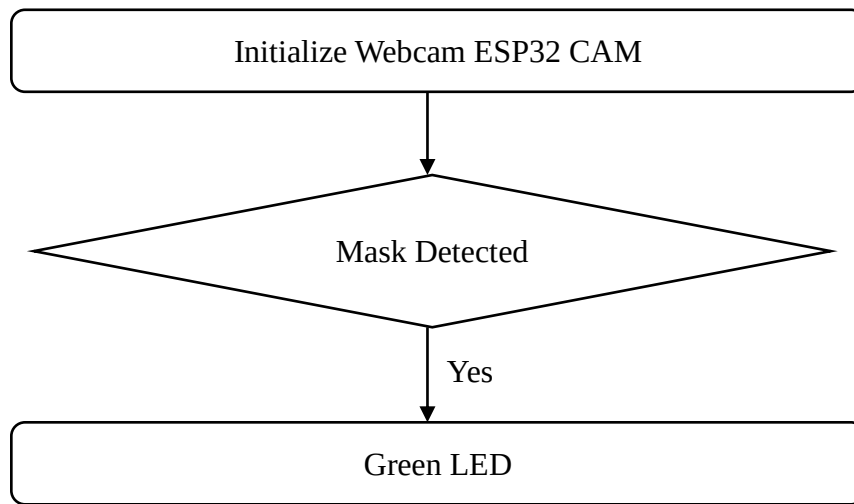


Fig. 6.2.1: Flowchart

First of all, we need to initialize the webcam i. e. ESP32 CAM module by giving power supply. Then the person needed to stand in front of the system. If mask is detected, green LED will glow.

7. CIRCUIT DIAGRAM

7.1 ESP32 CAM Programming Circuit

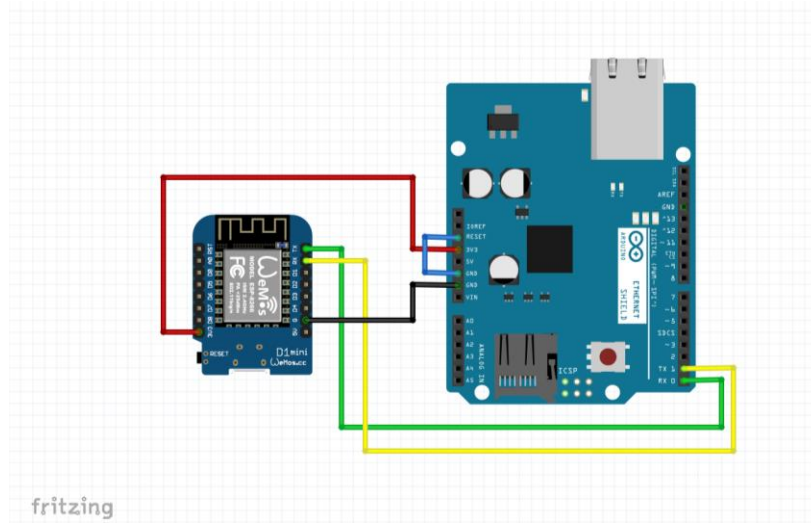


Fig. 7.1.1: ESP32 Programming Circuit

This circuit diagram is required for programming of ESP32 CAM module using Arduino Uno. Arduino is needed to connect the Arduino IDE software and then program is inserted by selecting ESP32 Wrover Module board.

7.2 Circuit Diagram

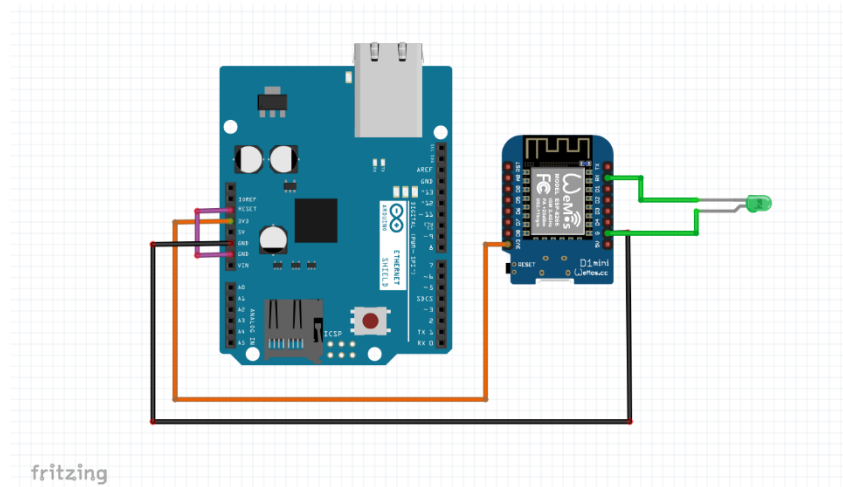


Fig. 7.2.1: Face Mask Detector Circuit Diagram

This is the circuit diagram of the system ‘Face Mask Detector’. A power supply is given to Arduino Uno by connecting adapter. Reset pin of Arduino Uno is connected to its ground pin. 3V3 pin of Arduino Uno is connected to 3V3 pin of ESP32 CAM module.

The green LED is connected to Rx pin of ESP32 CAM module. The cathodes of LEDs are connected to the ground terminal of ESP32 CAM module.

8. SELECTION OF COMPONENTS

8.1 ESP32 CAM Module



Fig. 8.1.1: ESP32 CAM Module

The ESP32-CAM is a small size, low power consumption camera module based on ESP32. It comes with an OV2640 camera and provides onboard TF card slot. The ESP32-CAM can be widely used in intelligent IoT applications such as wireless video monitoring, WiFi image upload, QR identification, and so on.

It supports WiFi with Bluetooth and comes with a OV2640 2MP Camera with Flash. It Supports multi sleep modes, deep sleep current as low as 6mA. Its Control interface is accessible via pin header, easy to be integrated and embedded into user products.

Its applications are smart home devices, image upload, video monitoring, smart agriculture, QR wireless identification, facial recognition, etc

8.2 Arduino Uno



Fig. 8.2.1: Arduino Uno

The Arduino Uno is an open-source microcontroller board based on the Microchip ATmega328P microcontroller and developed by Arduino.cc. The board is equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards (shields) and other circuits.

The board has 14 digital I/O pins (six capable of PWM output), 6 analog I/O pins, and is programmable with the Arduino IDE (Integrated Development Environment), via a type B USB cable.

The Arduino Uno is required in this system to program and provide a 3.3V supply to ESP32 CAM module.

9. SIMULATION

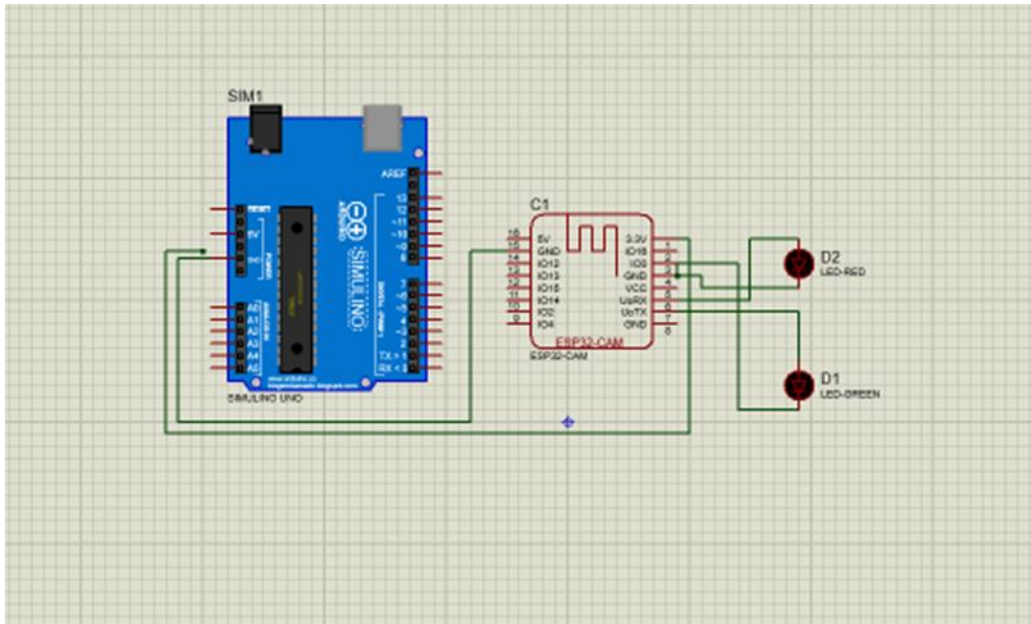


Fig. 9.1: Simulation Circuit

The simulation circuit of 'Face Mask Detector' system is designed using the Proteus Software. We cannot program ESP32 CAM module using proteus software, hence we used Arduino IDE compiler and tested the program using testing circuit.

10. TESTING PROCEDURE

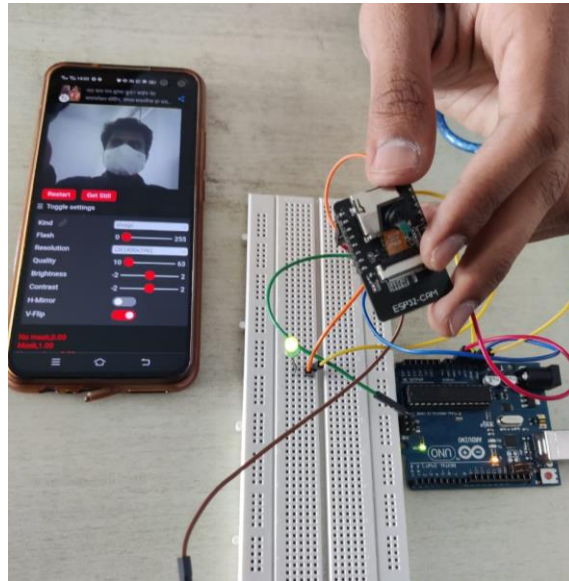


Fig. 10.1: Testing Circuit Using Breadboard

The testing of the circuit is done using a breadboard. The mask is detected and the green LED is glowing. From this we concluded that a person has worn face mask above the required percentage.

11. ENCLOSURE DESIGN



Fig. 11.1: Enclosure Design of 'Face Mask Detector'

The figure shows the enclosure design of our system 'Face Mask Detector'. It has a 2 MP camera and adapter and USB cable pin connector slots. Here, flash light is turned on and green LED is glowing because the person in front of the system has worn a face mask.

12. COST ESTIMATION

Sr. No.	Component Name	Price (Rs)
1.	Arduino Uno	860
2.	ESP32 CAM	790
3.	9V Battery	20
4.	Green LED	5
5.	Adapter	75
6.	Zero PCB	40
7.	Battery Connector	10
8.	Connecting Wires	40

Table. 12.1: Cost Estimation

Hence, total cost of this project is Rs. 1840.

13. CONCLUSION

Face Mask Detector system is very useful tool in this pandemic situation. It is difficult to check every person's face mask where public has gathered. This tough, tedious and high-risk task can be made easy by using this system.

It can be used in any type of building or place where public gathers in large number. It can be mounted on wall or placed in front of door of any room, so it will minimize the risk of infection as it separate outs people who are not wearing a mask. This system will definitely help us in this COVID situation.

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